

Brain-Computer Music Interface (BCMI): Exploring Creative Expression

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Abstract—Music has been shown to have a positive impact on human brain performance, with music stimuli commonly utilized in therapeutic applications to reduce anxiety, stress, and various mental disorders. This paper presents a novel Brain-Computer Music Interface (BCMI) system that can detect human brain activity, and convert them into different musical notes and visual elements. By mapping brain signals to both auditory and visual elements, the proposed system enables a unique and expressive music performance that is driven by brain activity.

To evaluate the performance and feasibility of the proposed BCMI system, it was tested with two distinct groups of participants: one with Music/Art background and the other without such background. The results suggest that the proposed system is practicable, and can potentially help people better communicate with each other.

I. INTRODUCTION

The human brain is a complex structure or network that contains 100 billion neurons [1] [2]. Human consciousness emerges from the processing of brain activity [2]. For many years, scientists have explored a variety of techniques for collecting and interpreting human brain activities. One of the most promising methods is a brain-computer interface (BCI) that enables individuals send messages or commands to the external world without passing through the brain's normal output pathways of peripheral nerves and muscles [3] [4] [5]. BCIs can be categorized into three modes: active, passive and reactive [6]. In the active mode, users can consciously control their brain activities without relying on external stimuli [6]. Active BCIs are commonly used for stroke rehabilitation [7]. In the passive mode, brain signals are generated unconsciously in response to external stimulation [8]. Passive BCIs are primarily used to predict users' mental and affective states [9]. In the reactive mode, brain activities are modulated in response to an external stimulus, thus making reactive BCIs well-suited for gaming and entertainment applications [10]. With BCI, people can use their consciousness as a medium and communicate with their origin thoughts, which has a potential to transcend the limitation of natural language.

A generic BCI system typically includes three components:

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- Signal acquisition system, which captures the electrical activity of a human brain in real-time. The brain signal can be collected by non-invasive methods such as electroencephalography (EEG) [11], or invasive methods such as Electrocorticography (ECoG) [12].
- Signal processing system, which analyzes the recorded brain signals and translates them into machine-readable commands.
- Output devices, such as a computer interface, a mobile device or a prosthesis [3] [13].

As affordable, portable and user-friendly EEG headsets (e.g., Muse, Emotive, and Brainlink) are available in the market, the BCI field has gained significant attention and interest. BCI has been applied in both clinical and non-clinical settings, such as medical rehabilitation [4], mental health medication [14], gaming [15], Virtual/Augmented Reality [16], and so on.

This paper demonstrates a novel Brain-Computer Music Interface (BCMI) system that enables performers to communicate through music and brain activities during their performances. The system involves two performers collaborating together, with one playing the piano while the other's brainwave activities are monitored using an EEG headset. The music produced by the piano serves as stimulus for the brain activities of the performer. The system processes real-time EEG data as input and maps them to visual and auditory outputs.

We aim to explore a constructive approach that represents brainwave data with musical/artistic expression, facilitating a deeper understanding of the relationship between human mind and creative performance. The remaining of the paper is organized as follows: Section II summarizes the previous studies on the effects music on human brain, and the concept of BCMI. Section III presents the system design, including EEG data acquisition and processing methods, as well as the sound and visual synthesis program. Lastly, Section IV and Section V contain the in-depth discussion, and conclusion of this study.

II. RELATED WORK

A. Music and Brain Activity

During the last decades, much research has been conducted on the effects of music on human brain. Previous studies have shown that infants and Toddlers are capable of processing musical information, indicating that human brain has the ability to respond and participate in music activity from a

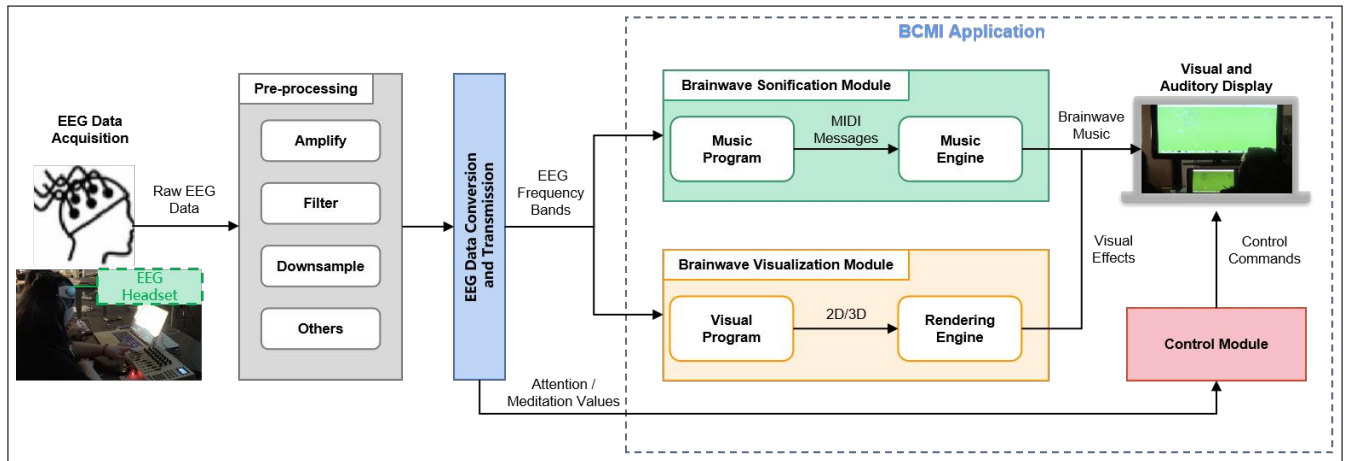


Fig. 1. System Overview: The proposed system includes EEG Data Acquisition, Pre-Processing, EEG Data Conversion and Transmission, Brainwave Sonification, Brainwave Visualization and Command Control.

very young age [17] [18] [19] [20]. For example, scientists Rock et al. observed that 6-month-old infants reacted differently to songs sung in a play-song style versus a lullaby style [17]. According to previous neuromusical research, musical processing is likely to be handled by different neural mechanism [21], and the human brain may react differently when hearing different types of music [22] [23]. For example, some studies indicate that classical music may increase the energy variation of alpha waves [23] [24]. A study by Saravananarajan et al. reveals that the wavelength of Theta and Beta waves increased on listening to peaceful music [24]. An experiment by Hassan et al. suggests that listening to live violin music may help in increasing balance between the left and right brain [25].

Furthermore, several studies have shown that music has a positive effect on the human brain performance [22] [25] [26]. Some studies also demonstrated that music may help to generate higher brain functional skills (e.g., reading, literacy, and mathematical abilities) [27] [28], improve long term memory [29] [30], and enhance brain plasticity [25]. However, Rahman et al.'s study suggests that music that induces patterns in the temporal and frontal lobes of the brain may have the potential to trigger seizures and therefore should be avoided [22].

B. Brain-Computer Music Interface (BCMI)

A Brain-Computer Music Interface (BCMI) is a type of BCI system that enables a user to voluntarily controls music [31]. In 2008, Miranda et al. first introduced a proof-of-concept system called BCMI-Piano that used EEG information to compose music online, which played on a MIDI-enabled acoustic piano [32]. Later, Eaton et al. demonstrated a real-time hybrid BCMI that combined both active and passive BCI control [33]. Furthermore, Eaton et al. presented a BCMI that could measure the affective states of a performer and an audience member during a live musical performance [34] [35]. Tokunaga and Lyon also demonstrated a BCMI prototype that could reflect a user's emotional and cognitive state [36]. Tokunaga et al.'s system utilized real-time EEG

data to generate kinetic audio-visual representations and artificial expressions, reflecting a user's physiological states. According to previous researches, there are three major challenges for developing a BCMI [31]:

- Extracting meaningful control information from brain signals.
- Creating generative music techniques that can respond to such information.
- Investigating effective ways in which this technology can enhance the lives of individuals with special needs and address therapeutic needs.

III. PROPOSED METHODOLOGY

In the proposed system (see Fig.1), we used a consumer level EEG headset with our signal processing program coupled with the musical instrument data interface (MIDI). The sound and visual synthesis program were implemented using Ableton Live 10 and Processing software environments. The proposed system includes six modules: EEG Data Acquisition module for real-time monitoring of the user's brainwave data (EEG); Pre-Processing module for pre-processing the raw EEG data; EEG Data Conversion and Transmission module for converting EEG data to a supported data format; Brainwave Sonification module for facilitating music composition; Brainwave Visualization module for generating visual elements and rendering the visual scenes accordingly; Command Control module for triggering corresponding commands.

A. Pre-Experiment

The pre-experiment was implemented for exploring the brain responses to the music perception. During the experiment, a participant was instructed to wear an EEG headset and listen to various genres of music (see Fig.2), and brainwave visualization software was utilized to monitor the participant's brain activity in response to each music track (see Fig.3). It was observed that the participant's brain rhythms synchronized with the musical sequences and tempo. Specifically, soft music elicited a smooth pattern in the brainwaves,

while energetic music led to more erratic brainwave patterns. This observation aligns with a study by Doelling et al., which demonstrated that cortical oscillatory activity in both low (<8 Hz) and high (15–30 Hz) frequencies is closely linked to behavioral performance in musical listening, in a bidirectional manner [37]. The pre-experiment provided the insight that the frequency of brainwave could be represented by musical elements as they correspond with each other. A video demonstrating the pre-experiment process is available (refer to Video: <https://github.com/karima931212/Brainwave-Instrument/blob/master/experiment.mov>).

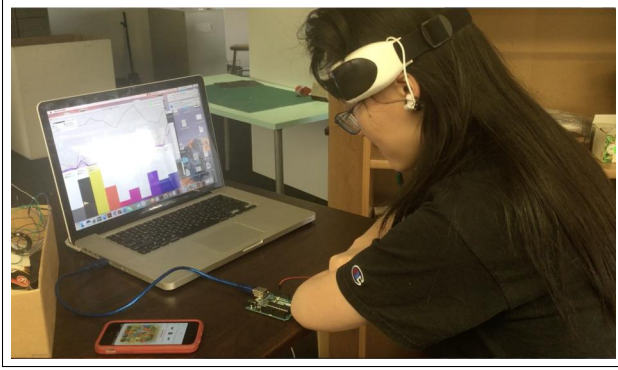


Fig. 2. Experiment process: a participant was listening to music and there was a EEG device monitoring her brain activities in real time.

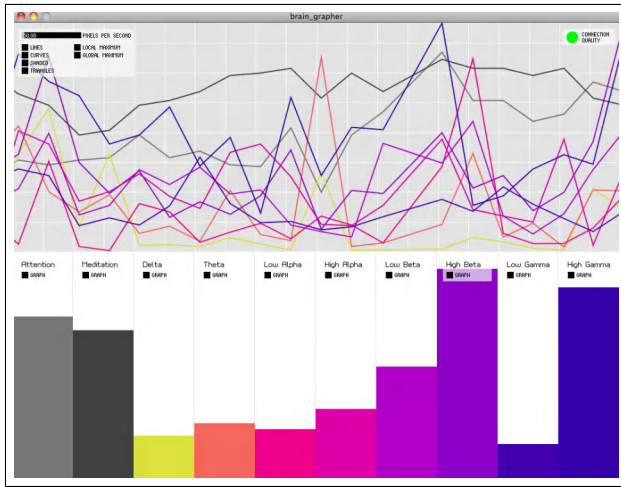


Fig. 3. Brain Activity Monitor.

B. EEG Data Acquisition

The EEG data were collected through a non-invasive EEG headset, Mindflex. It detected brain signals through a metal electrode attached to the forehead and two reference electrodes placed at left and right ear mastoids via the monopolar method.

C. Pre-Processing

Pre-processing of the collected raw EEG data included tasks such as amplification, filtering, and downsampling. These can be performed by the brainwave monitoring device

itself or by additional algorithmic modules such as MNE-Python. For example, the Mindflex headset comes with built-in pre-processing functionality. The processed EEG data was presented in the form of byte streams, which include 8 brainwave frequency bands as well as measures of attention and meditation, as shown in Table 1.

TABLE I
EEG CHANNELS AND FREQUENCY EQUIVALENTS [38]

Brainwave Type	Frequency Range	Mental States
Delta	1-3Hz	Deep sleep Unconscious
Theta	4-7Hz	Creativity Deeply relaxed Daydreaming Meditative
Low Alpha	8-9Hz	Mental coordination Passive attention Calmness Alertness
High Alpha	10-12Hz	
Low Beta	13-17Hz	Alert Focused Attentive Active thinking
High Beta	18-30Hz	
Low Gamma	31-40Hz	Concentration Cognitive functioning
High Gamma	41-50Hz	
Attention	0-100	Indicates the intensity of a user's mental "concentration" or "attention" level.
Meditation	0-100	Indicates the level of mental "calm" or "relaxed" of the user.

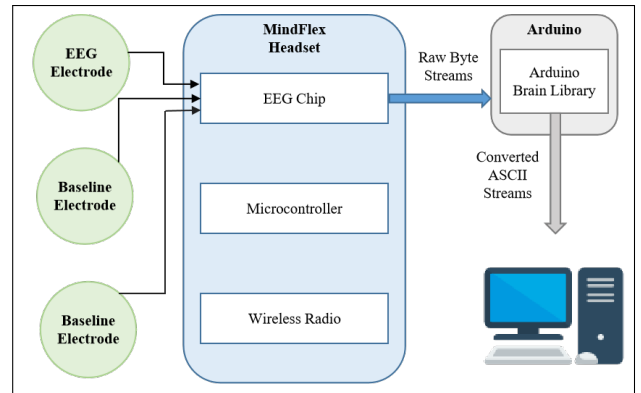


Fig. 4. EEG Data Conversion Process.

D. EEG Data Conversion and Transmission

The pre-processed EEG data was initially in the form of a byte stream. We then converted this byte stream into an ASCII stream using Arduino Brain Library, and sent the converted stream to a PC via a USB interface. The conversion process is summarized in Fig.4.

In order to transmit the converted EEG data to various software applications, such as Pure Data and Processing, a multicast Open Sound Control (OSC) protocol server was established. OSC is a protocol for communication among computers, sound synthesizers, and other multimedia devices that is optimized for modern networking technology. It offers high accuracy, low latency, and flexible data transport capabilities, making it an ideal method for real-time message communication among applications and hardware components.

E. Brainwave Sonification

A music program was developed using Pure Data to unpack the acquired EEG stream, extract the EEG data and convert the EEG data into MIDI messages (see Fig.5). The MIDI messages generated by the music program were then used to manipulate the performance of the virtual instruments in the music engine created by Ableton Live. For this, we first created a MIDI Track in the music engine for each EEG frequency band, with a specific virtual instrument assigned to each track (see Fig.6). For instance, the low alpha band was used to control an electronic guitar.

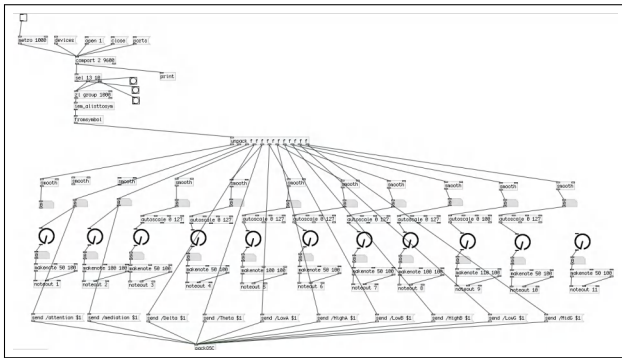


Fig. 5. Music Program converted EEG data into MIDI messages (range from 0 to 127).



Fig. 6. Music Engine developed by Ableton Live.

The general steps are as follows:

- Using an “unpack” object, the EEG stream was separated into eight individual channels, each corresponding to a specific EEG frequency band.

- For each channel, the EEG data was extracted and mapped to MIDI messages ranging from 0 to 127 using the “autoscale” object.
- The generated MIDI messages were then sent in real-time to Ableton Live music engine using a “noteout” object. This allowed for the triggering of music composition and performance based on the EEG signals.

F. Brainwave Visualization

To add a visual element to the performance, the EEG data were simultaneously transmitted to a visual program developed using Processing software via the OSC protocol. This program was designed for triggering synchronized visual effects that complemented the music. Abstract shapes were created to represent different types of brainwaves, with each shape dynamically changing in size according to the frequency of the corresponding brainwave. These visual elements were used as cues for the two performers to communicate with each other during the performance, enhancing their collaboration and expression.

G. Command Control

The user’s attention and meditation levels were used for command control. When these values surpassed a predefined threshold, corresponding commands were activated in the application. For instance, the attention level was measured on a scale from 0 to 100, and based on our experimental findings, it had been observed that when the user was highly focused with an attention level exceeding 60, a command to change the background color was automatically executed in the application.

IV. DISCUSSION

The proposed system was tested with two distinct groups of participants: one comprising individuals with Music/Art background and the other without such background. Both groups were given a brief instruction on how to use the system, and allowed to familiarize themselves with the music and visual generation process. However, they were not informed about the relationship between the visuals, music, and brain activity.

Both groups were able to recognize that the music was generated by the performer’s brain. The Music/Art group initially experienced confusion, but quickly recognized the correlation between visual changes and the music. One participant remarked, “I noticed that the shapes and sound were related, changing at the same time.” As they became more familiar with the system, they were able to comprehend that the visual elements represented the performer’s brain activity.

In contrast, the group without Art/Design background had difficulty understanding the visual elements and communication with abstract shapes was unclear for them, with one participant stating, “I noticed that the sound became very sharp when I was excited or talking, but the shapes were just circles, triangles, and squares, so I didn’t relate them to my brain activity.”

V. CONCLUSION AND FUTURE WORK

In conclusion, this paper demonstrated a real-time BCMI system which was built with consumer-level hardware and open-source software. By bridging the fields of neuroscience and artistic expression, the proposed system effectively translated captured EEG data into visual and auditory elements, which were used as communication cues. The results highlight the potential of sonifications or visualization of brain signals to create a unique musical experience and facilitate improved interpersonal communication.

For the future work, more mapping methods could be investigated to better interpret brain signals. In addition, a comprehensive brainwave visualizer interface needs to be incorporated. With advancements in neuroscience, EEG headsets and signal processing algorithms, we expect that the BCMI system will be more scientific, and enabling direct brain communication.

The Brainwave Virtual Instrument can be found at <https://github.com/karima931212/Brainwave-Instrument> including Pure Data, Adurino, Processing code and Ableton Live project. The video demonstration can be found at <https://vimeo.com/271557587>.

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